

LG Vehicles Report Workflow

Generates a vehicle movement analysis report for Lion Guardians in the Amboseli ecosystem, sourced from EarthRanger.

What it produces

The workflow produces, for each tracked vehicle:

- A **Speed Map** — trajectory segments coloured by a 6-class speed palette (green → red)
- A **Vehicle Tracks Map** — uniform blue path overlay on the study area layers
- A **Speed Over Time line chart** — speed (km/h) plotted against date per vehicle
- **Scalar metric widgets** — Mean Speed, Min Speed, Max Speed, Distance covered
- A **per-vehicle summary CSV** — speed and distance statistics (one row per vehicle)
- A **Word document report** (`.docx`) — cover page plus one section per vehicle
- An **interactive widget dashboard**

Requirements

- Access to an **EarthRanger** instance with a configured data source
- The **Vehicles** subject group present in your EarthRanger instance
- The **Conservancies** and **Group Ranch Boundaries** spatial features must exist in that EarthRanger instance — these are fetched live to build the study-area map layers

Both Word report templates and the organisation logo are downloaded automatically from Dropbox — no local copies are required.

1. Load the Workflow

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste this repository's URL into the **Github Link** field, then click **Add Template**:

```
https://github.com/wildlife-dynamics/lg-vehicles-report.git
```

Once added, it appears in the **Workflow Templates** list as **lg-vehicles-report**. Click it to open the workflow configuration form.

The card may show **Initializing...** briefly while the environment is set up.

2. Configure the Workflow

Data Source Connection

Navigate to **Data Sources** and add a new EarthRanger connection. Fill in:

- **Data Source Name** — a label to identify this connection
- **EarthRanger URL** — your instance URL (e.g. `your-site.pamdas.org`)
- **EarthRanger Username** and **EarthRanger Password**

Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Workflow Details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes about the run (e.g. date range or vehicle group)

Time Range

Field	Description
Timezone	Select the local timezone (e.g. <code>Africa/Nairobi UTC+03:00</code>)
Since	Start date and time of the analysis period
Until	End date and time of the analysis period

All vehicle tracks and metrics are computed within this window.

Basemap Layers

Two stacked ArcGIS tile layers form the background of every map. Pre-filled with sensible defaults, but the URL, opacity, and max zoom of each layer are editable.

Layer	Default Opacity	Max Zoom
ESRI World Hillshade	1.0	15
ESRI World Street Map	0.15	15

Connect to EarthRanger

Select the EarthRanger connection configured above from the **Connect to EarthRanger** dropdown. The workflow will fetch all observations for the **Vehicles** subject group from this instance.

Trajectory Filter

Expand **Advanced Configurations** under **Transform relocations to trajectories** to adjust the segment filters. These parameters control how raw GPS fixes are converted into trajectory segments.

Field	Default	Description
Minimum Segment Length (m)	3	Discard segments shorter than this distance
Maximum Segment Length (m)	100000	Discard segments longer than this distance
Minimum Segment Duration (s)	1	Discard segments shorter than this duration
Maximum Segment Duration (s)	21600	Discard segments longer than this duration (6 hours)
Minimum Segment Speed (km/h)	3	Discard segments below this average speed
Maximum Segment Speed (km/h)	150	Discard segments above this average speed

Zoom to Envelope

Field	Default	Description
Expansion Factor	1.05	Factor to expand the bounding box when auto-zooming maps (e.g. 1.2 = 20% larger)

3. Run the Workflow

Once all parameters are configured, click **Submit**. The runner will:

1. Pull vehicle GPS observations from EarthRanger for the specified time range.
2. Fetch the Conservancies and Group Ranch Boundaries spatial features from EarthRanger to build the study-area map layers.
3. Convert observations to relocations, then build trajectory segments with speed and distance metrics.
4. Classify speed into 6 equal-interval bins and generate the Speed Map.
5. Render the Vehicle Tracks map using a uniform blue path layer.
6. Draw the speed-over-time line chart per vehicle.
7. Compute per-vehicle summary statistics (mean, min, max speed; total distance).
8. Assemble the Word report (cover page + per-vehicle sections) and the dashboard.
9. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS` .

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/` :

File	Description
<code>vehicle_relocations.geoparquet</code>	Cleaned GPS fix locations
<code>vehicle_trajectories.geoparquet</code>	Trajectory segments with speed and distance
<code><vehicle>_speedmap.html</code>	Interactive speed map per vehicle
<code><vehicle>_tracks.html</code>	Interactive tracks map per vehicle
<code><vehicle>_speed_line_chart.html</code>	Interactive speed-over-time line chart
<code><vehicle>_speedmap.png</code>	Screenshot of speed map (2× resolution)
<code><vehicle>_tracks.png</code>	Screenshot of tracks map (2× resolution)
<code><vehicle>_speed_line_chart.png</code>	Screenshot of speed line chart (2× resolution)
<code><vehicle>_summary.csv</code>	Speed and distance summary table (one row per vehicle)
<code>cover_page.docx</code>	Rendered report cover page
<code><vehicle>.docx</code>	Per-vehicle report section
<code>overall_report.docx</code>	Final combined Word report

More Help

- **Technical Guide:** [technical_guide/lg_vehicles_report_technical_guide.pdf](#) — pipeline internals and task-by-task reference
- **Issues:** github.com/wildlife-dynamics/lg-vehicles-report/issues

Development

This workflow's code (`ecoscope-workflows-vehicles-report-workflow/`) is generated from [spec.yaml](#) and [test-cases.yaml](#) . After editing either file, recompile and commit the generated changes:

```
pixi run --manifest-path pixi.toml --locked bash -c "./dev/recompile.sh --update"
```

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